

Virtual Playing, Real Touch: The Haptics of Musical Instruments

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Abstract

Virtual reality (VR) has been used as a medium for interaction with objects and environments, yet hand-based interaction in VR often lacks physical contact and resistance. Despite the recognized importance of physical touch, it remains unclear to what extent virtual haptic feedback can replicate real-world tactile experiences. In this work, by using HaptX haptic gloves, we developed an interactive haptic VR system where users can experience three types of haptic interaction (plucking, pressing and patting) through playing multimodal musical instruments with realistic visual, auditory and haptic feedback. Haptic feedback can be switched on or off while keeping tracking and audiovisual rendering constant, enabling direct comparison of the interactions with and without touch cues. The demo highlights how the perceived role of touch can differ across action types and provides a platform for exploring when tactile and resistive feedback meaningfully changes VR interaction.

CCS Concepts

• **Human-centered computing** → **Haptic devices**.

Keywords

Virtual reality, Multisensory experience, Haptic feedback, User experience

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1 Introduction and Related Work

Physical musical instruments offer rich and expressive interaction, but they also introduce practical constraints. Instruments can be

bulky, expensive, and acoustically intrusive, making regular practice difficult in shared or space-limited environments [16]. Virtual reality (VR) has therefore been explored as an alternative medium for musical interaction and practice, offering greater flexibility, privacy, and control over audiovisual conditions [7]. Prior work has investigated how immersive environments can support creative musical practice, while also highlighting that the quality of interaction plays a central role in whether such systems feel meaningful or merely novel [14].

Most VR-based musical systems rely on handheld controllers, which provide robust tracking, but they lack finger-level interaction [7]. While this simplifies input, it does not capture the fine-grained hand movements that characterize many musical actions. In response, many systems use controller-free hand tracking, but it often trades reliability for expressiveness and still provides no physical contact or resistance [2]. Comparative studies of VR input modalities further suggest that the choice between controllers and hands can significantly affect workload, perceived control, and collaboration, even when task goals remain unchanged [1]. For instrument-like interactions that depend on timing, contact, and subtle force modulation, the absence of haptic feedback becomes especially important [17].

To implement touch sensation, prior research has used a wide range of haptic devices for hands and fingertips, highlighting the trade-offs between wearability, force magnitude, spatial resolution, and mechanical complexity [13]. At the interaction level, studies have shown that richer fingertip feedback can meaningfully influence both perception and action during virtual object interaction, affecting movement strategies and object recognition [17]. Recent wearable gloves such as SenseGlove¹ and HaptX² illustrate how touch can be reintroduced into hand-based VR through combinations of finger tracking and physical feedback (for example, contact cues and resistive forces) [8, 15]. As such, our demo uses *HaptX DK2* to provide both tactile cues at the hand and resistive forces at the fingers, enabling attendees to compare interaction with haptic feedback enabled versus disabled while keeping visuals and audio constant.

Adding touch to hand-based VR interaction can change input fidelity and the felt experience of contact, effort, and timing during



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¹<https://www.senseglove.com/>

²<https://haptx.com/>



Figure 1: User wearing HaptX DK2 gloves and an HTC Vive headset. The haptic feedback can be enabled or disabled while keeping tracking, visuals, and sound constant, supporting direct comparison of interaction with and without touch cues.

action. These aspects are not always captured well by accuracy-style measures, and prior work argues for treating bodily engagement and felt experience as central to interaction design [6, 10]. Building on this perspective, our interactive demo invites attendees to compare the same musical interactions with haptic feedback enabled versus disabled.

We previously evaluated the system in a controlled study (N=105) across three conditions (VR with haptics, VR without haptics, and physical instruments), and the demo presented here allows attendees to experience the contrasts between VR with haptic feedback and VR without haptic feedback hands-on (Figure 1).

2 Design Strategy

Our design strategy is built around **contrast rather than optimization**. The demo is intended to make the presence or absence of touch noticeable without changing other aspects of the experience. To do this, we keep the virtual environment, visuals, and sound rendering constant while switching haptic feedback on or off. Similar contrastive approaches have been used to highlight experiential differences that are otherwise difficult to articulate, particularly for embodied and multisensory interaction [5, 6].

To focus attention on how touch changes interaction in VR, each variant is structured around a simple action type that is common in instrument playing and everyday hand use. The three actions differ in contact duration and force profile: sustained pressing, brief release-based interaction, and percussive impact. Using these simple actions helps clarify how tactile and resistive feedback can shape perceived effort, control, and timing during interaction [3, 12].



Figure 2: The user presses the virtual keys using downward finger motion and sustained contact (left) on the Piano variant in VR (right).

The demo supports two configurations: (1) visual and auditory feedback only, and (2) the same interaction augmented with tactile and resistive feedback delivered through *HaptX DK2* gloves. In both configurations, users wear the same gloves to ensure identical hand tracking. In the non-haptic mode, haptic feedback is disabled while tracking, visuals, and audio remain unchanged. The gloves provide localized tactile cues and finger resistance during contact, supporting physically grounded interaction that goes beyond vibration-only feedback. Related work has explored combinations of tactile and kinesthetic feedback to enrich hand-based interaction in virtual environments [4, 11].

Rather than asking whether haptics is “better,” the demo invites attendees to notice how touch changes specific aspects of interaction across the three variants. For example, haptics can affect the sensed “key travel” during pressing, the moment of release during plucking, or the sharpness of impact during percussive contact. This aligns with prior work arguing that interactive systems can support reflection and design insight by making bodily and felt experience more available to the user [9, 10].

3 Technical Implementation

The demo is implemented as a seated VR experience system using an HTC Vive headset and *HaptX DK2* haptic gloves. The experience runs in Unity 2021.3 LTS on a high-performance laptop, and users interact with the instruments using glove-based hand and finger tracking (no handheld controllers).

3.1 Sensory Modalities Implementation

Haptics: The HaptX DK2 gloves are driven via HaptX SDK 2.1.0 and provide two types of haptic feedback: pneumatic tactile cues delivered to the fingertips and palm, and resistive forces that constrain finger motion during contact. Haptic feedback through the gloves can be enabled or disabled while keeping tracking unchanged, supporting direct comparison of interaction with and without haptic cues.

Visual: The three instruments were modeled in Blender to closely match the geometry and proportions of the corresponding physical instruments used in our setup. We experimented with an Artec 3D scanning workflow to obtain reference geometry, but the final instrument meshes were created and refined in Blender, with particular attention to visible features and finger contact regions. The interaction states (e.g., key travel, string displacement, surface contact) are visually animated to match user input.

Audio: Each instrument produces immediate sound responses to finger contact events. Audio playback is synchronized with visual

events and haptic rendering to reduce perceptual mismatch across modes.

3.2 Instrument Interaction Implementation

Piano (pressing): Each key is represented as an articulated key mechanism using a Unity HingeJoint. A note is triggered when the key rotates past a press threshold, analogous to reaching a key-down point during a physical press. To avoid repeated triggering while the key is held, the event is re-armed only after the key returns near its rest position.

Guqin (plucking): Each string is implemented as a trigger region that detects fingertip contact. A pluck is recognized when the fingertip motion includes sufficient velocity perpendicular to the string direction, approximating the lateral pull that precedes release in physical string plucking. On a valid pluck, the system plays an audio response and triggers a brief visual vibration animation.

Log drum (patting): Each drum surface is implemented as a collision-based pad that triggers a note when a fingertip impacts the surface above a minimum velocity threshold, approximating a discrete strike.

Across all variants, the same hardware and software pipeline is used. The system is designed for repeatable hands-on sessions, allowing users to switch between modes while keeping visuals and audio constant.

4 Demo Variants

The demo consists of three instrument variants that span different hand contact profiles. In each variant, haptic feedback can be toggled on or off, enabling direct comparison of the same interaction with and without touch cues while keeping tracking, visuals, and audio unchanged.

4.1 Playing Piano - Pressing

The piano involves sustained fingertip contact while pressing keys (Figure 2). Each key provides immediate visual motion and sound feedback on press. With haptic feedback disabled, interaction is guided primarily by audiovisual cues. With haptic feedback enabled, users additionally feel fingertip touch cues and resistance during key presses, which can change the perceived weight of the press and the timing of release.

4.2 Playing Guqin - Plucking

The guqin requires brief pull-and-release actions to pluck strings (Figure 3). Without haptic feedback, users rely on visual timing and sound cues to judge contact and release. With haptic feedback enabled, brief touch cues accompany string interaction, making the moment of release and the sharpness of the pluck easier to perceive.

4.3 Playing Log Drum - Patting

The log drum consists of short percussive strikes on discrete surfaces (Figure 4). Without haptic feedback, strikes are mediated by visual impact and sound. With haptic feedback enabled, users receive impact-like tactile cues and brief resistance during contact, which can affect the perceived sharpness and physicality of each hit, especially during rapid patting.



Figure 3: The user plucks the strings using lateral finger motion (left) on the Guqin variant in VR (right).



Figure 4: The user performs short percussive strikes (left) on the Log Drum variant in VR (right).

5 Demo Booth

At the demo booth, attendees will try each instrument variant and can switch haptic feedback on/off to compare interaction modes directly. A mirrored display will show the virtual scene to observers. The experience is designed for short sessions with quick turnover, with basic cleaning between users for shared wearable components.

6 Discussion and Conclusion

This work presents an interactive VR demo that lets attendees compare the same musical interactions with haptic feedback through the gloves enabled or disabled. Across three instrument variants, the demo highlights how adding touch cues can change perceived control and timing even when visuals and sound remain constant.

The design emphasizes contrast and simplicity, allowing users to understand the changes in effort, control, and timing without added complexity from the task or the need for extended learning. Across the three variants, the demo is designed to make it easy to notice that the role of touch can differ by action type, suggesting that haptics may matter in different ways for pressing, plucking, and percussive contact. These observations reinforce prior arguments that bodily experience and effort are central components of interaction, and that their effects are not easily reduced to performance metrics or accuracy measures [17].

This work treats haptic interaction as a hands-on comparison rather than a formal evaluation. By allowing users to directly experience differences between haptic and non-haptic interaction, the system provides a concrete basis for discussing when and how touch may be meaningful in virtual musical contexts.

As an interactive demo, the system is designed to be robust, repeatable, and scalable to different musical instruments. The interactions are short, self-contained, and require minimal explanation, enabling hands-on exploration without specialized expertise. This work encourages further discussion around the nuanced role of haptic feedback in multisensory VR interaction, and supports future research and design efforts that treat touch not as a binary feature, but as a rich and context-dependent component of experience.

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